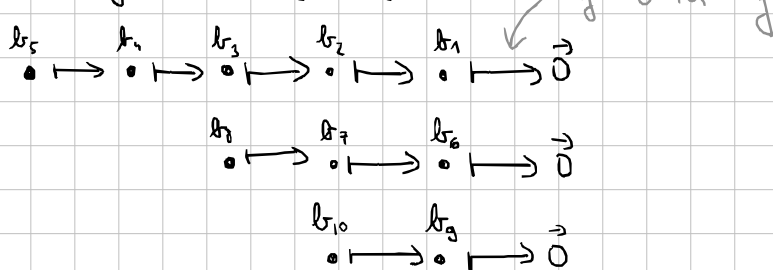


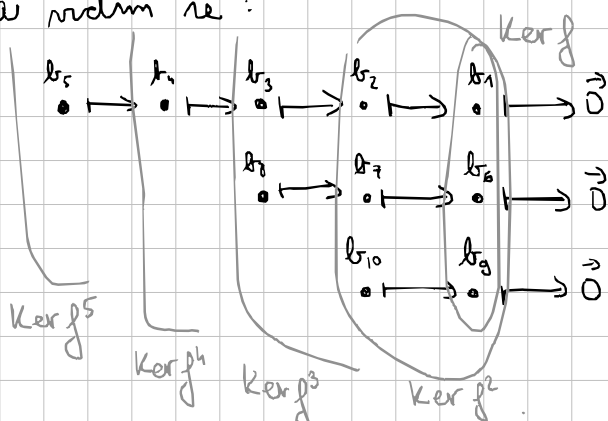
7.1. $f: \mathbb{R}^{10} \rightarrow \mathbb{R}^{10}, B = (b_1 \dots b_{10})$

$$J = [f]_B = \begin{pmatrix} \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}} & & & \\ & \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}} & & \\ & & \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}} & \\ & & & \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}} \end{pmatrix}$$

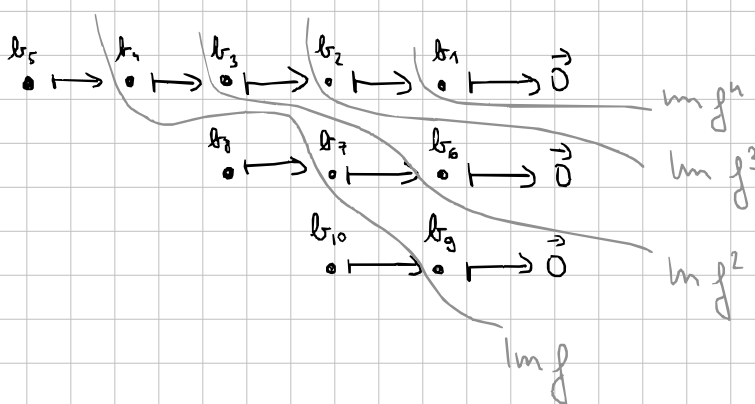
keď riešiky vypadajú:



a vidíme že:

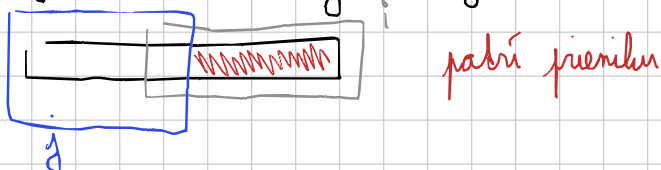


$\text{Ker } f^i$ je vždy prvých i vektorov z každého riešiku



$\text{Im } f^j$ je vždy každý riešik bez posledných j vektorov

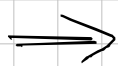
keď pre jednotlivé riešiky / linky:



pre linku $k \times k$: $\dim \text{Ker } f^i = \min \{i, k\} = x$
 (a f omerané na tú linku) $\text{Ker } f^i = \text{Span} \{b_1 \dots b_x\}$

$\dim \text{Im } f^j = \max \{0, k-j\} = y$ (ak by malá matica prebrala ďalšie riešiky)
 $\text{Im } f^j = \text{Span} \{b_{x+1} \dots b_k\}$

$\dim \text{Ker } f^i \cap \text{Im } f^j = \max \{0, \min \{i, k-j\}\} \leftarrow (f \cap \text{linka})$



$\dim \text{Ker } f^i \cap \text{Im } f^j = \max \{0, \min \{i, 5-j\}\} + \max \{0, \min \{i, 3-j\}\} + \max \{0, \min \{i, 2-j\}\}$

$\text{Ker } f^i \cap \text{Im } f^j = \text{Span} \{b_1 \dots b_x, b_{x+y+1}, b_{x+y+2}, \dots, b_{x+y+z}\}$

(pokiaľ y alebo z bude 0, tak neberem z daného riešiku nič)

(pokiaľ y resp. z bude 1, tak berem len b_{x+y} resp. b_{x+y+1} a nič ďalšie nepridávam)